

Colour Image Recognition using a Novel Variant of Convolutional Neural Network

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Abstract: Numeral recognition is an important preliminary step for optical character recognition (OCR), document understanding and others. Deep learning based numeral recognition models have achieved tremendous popularity among the researchers around the world since the last few years. Several convolutional neural networks (CNN), which are based on architectures have been proposed and many of those have achieved state-of-the-art results in numeral recognition. In this paper, we have explored CNN based architectures for colour image(cat-dog) recognition. We have also developed a colour image (cat-dog) dataset using various morphological operations on an existing dataset – thus increasing the size of the dataset from 2000 to 10000 images. A modification of previously proposed CNN architectures has given us an accuracy of 92.00% and our proposed architecture has produced 93.12%, which is comparable to state-of-the-art results found in the domain of colour image (cat-dog).

Keywords: OCR, Deep learning, CNN, Handwritten numeral recognition and Colour image(cat-dog).

1.Introduction.

Convolutional Neural Networks are often called ConvNets, which is kind of Neural Network Architecture that are mostly used in Image Classification and ConvNets are very good when it comes to Images. ConvNets was inspired by the Human Visual Cortex. We try to build a ConvNet in this paper. The Dogs vs. Cats dataset is a standard computer vision dataset that involves classifying photos as either containing a dog or cat. Although the problem sounds simple, it was only effectively addressed in the last few years using deep learning convolutional neural networks. Although the dataset is effectively resolved, it can be used as the basis for learning and practice in how to develop, evaluate, and use convolutionally deep learning neural networks for image classification from scratch. These include how to develop a powerful test harness for estimating a model's performance, how to explore the model's improvements and how to save the model and how to load it to make predictions on new data. You will discover how to develop a convolutional neural network for classifying photos of dogs and cats. colour image recognition is an active area of research in the domain of pattern recognition. It is an important subfield of Optical Characters Recognition (OCR). In the field of automatic document processing and analysis, the correct interpretation of numerals and characters are very important as it has wide commercial importance. Numeral recognition, both handwritten and printed, is used for several purposes like in post offices for scanning zip code to arrange mails, in the colour image for processing, image plate, etc.

Automatic recognition of colour image is a challenging task due to several reasons such as huge variation in the image style of different individuals, Due to all these difficulties, the development of an accurate colour image recognition system is still considered as an active area of research .Domain such as MNIST [2], CENPARMI [4], ARDIS [5] etc.

2. Related Work:

Pattern recognition is considered as one of the most important and challenging research domains. Many works have been done on different pattern recognition problems. One of the important sub-domains of this is colour image recognition, which is a required component of any document image processing system or the image understanding system. Some of the recent research works performed by the others on colour image recognition for the kaggle dataset are briefly discussed here. The dog vs. cat competition from kaggle is trying to solve the captcha [2] challenge, which depends on the problem of separating the image of dogs and cats .This is easy for humans, but evidence [2] suggests that cats and dogs are difficult to separate automatically .Many people have been working or working to create machine learning classifiers to solve this problem. In [2], a classifier based on colour properties found

56.9% accuracy in the Asirra dataset [3]. In [6], 82.7% accuracy was obtained from SVM classification based on a combination of colour and texture properties. And, in [7], they used SIFT (scale-invariant feature transform) [1] features to train a classifier and finish. Up to 92.9% accuracy was achieved. In our project we also want to solve this problem and achieve higher performance. We have tried different techniques. For example, we tried the combination of thicker SIFT features and thicker SIFT and colour features and learned features from CNN. Additionally, we employed SVMs on the learned features and finally achieved 93.12% accuracy to our best classification.

3. Experimental Setup:

Our main task is to create an algorithm to classify whether a dog or a cat is in a picture. The input to this task is the image of the dog or cat training dataset, while the output is the accuracy of the classification of the test dataset. The dataset provided for this competition is the Kaggle dataset provided by Microsoft Research. Our training set contains 10,000 images with 5000 images of dogs and 5000 images of cats, while the test dataset contains 2,000 images. The average size for these images is about 150×150 . Our task is to learn a classification model to determine the boundaries of the training dataset's decision. We see the images in the training dataset for the learned task, and the output is the learned classification model.

4. Proposed Methodology:

In recent times, CNNs have been used extensively for character and image recognition tasks, which have proven to be most effective model in terms of performance for multi-script image recognition task. Deep neural network's recent progress has tremendously promoted the result of Colour image recognition. CNN based framework outperforms previous classical approaches using handcrafted features for image recognition work. CNN is a multi-layered hierarchical model which is robust enough for character recognition work having real-world constraints. An intuitive explanation of basic layers of CNN model is discussed as follows.

Convolution Layer:

The primary focus of this layer is to extract high level spatial features from input images using convolutional approach and generate feature maps. Let, be an input image and is the filter then, generated feature map. These kernels/filters are usually small in dimensionality compared to input image, but convolve across the depth of the image and generate activation map or feature map. In convolution layer different combinations of kernel is used to extract different spatial features like horizontal and vertical edge detection, corner point detection, Gaussian blur etc. The size of the feature map depends on three parameters:

Depth:

Number of different kernels is used in convolutional layer defined as depth of feature map.

Stride:

Stride defines as number of pixels by which kernel slides over the input image. If, stride size set is 1, it means pixel by pixel convolution is performed. Increase the stride size leads to a small dimensional feature map.

Zero padding:

Adding fuzzy zero padding around the boundary of the input image covers boundary pixels in the convolution operation. This technique gives further control to the dimensionality of the output feature map. Fig. 1. illustrates the basic operation of convolution layer.

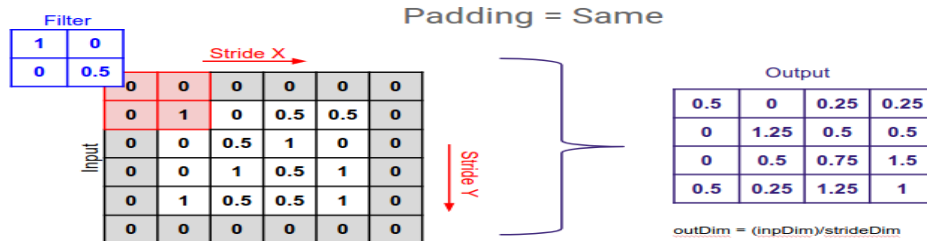


Fig. 1. Graphical representation of convolution operation, the center pixel of kernel is placed over the input image and performs dot product to generate corresponding feature map.

Activation Layer:

This layer increases the non-linearity of the model without disturbing the receptive elements of previously generated feature map. Several activation functions are used in CNN, but Rectified Linear Unit (ReLU) is the most preferred one. ReLU is used just after convolution layer and before subsampling layer, where all negative values of feature map are replaced with zero. The function of ReLU is:

Other non-linear activation functions like tanh, sigmoid etc. may be used instead of ReLU in CNN.

Sub-sampling/Pooling Layer:

Sub-sampling layers aim to gradually reduce the feature map dimensionality but it retains most important spatial information which leads to compute low cost of the model. Pooling layer operates on the feature map and reduces the dimension by using different pooling functions like “max pooling” or “average pooling”. In max pooling operation, it reduces the dimension of feature map by pooling out the maximum value within a defined pooling window. In “average pooling” operation, it pools out the mean of all values of feature map within the defined pooling window. Fig. 2 shows the pooling operation of sub-sampling layers.

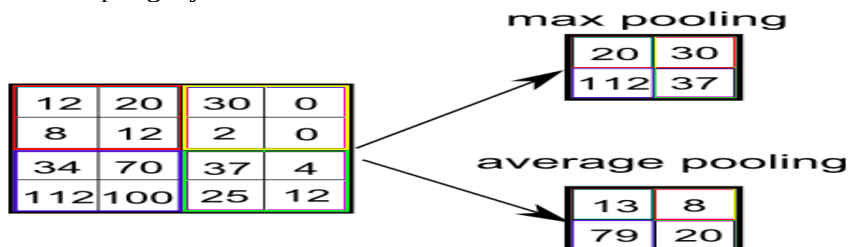


Fig.2. Visual representation of pooling operations, (a) max pooling, (b) average pooling. In most cases max pooling is used for CNN architecture.

Fully Connected Layer:

In Fully Connected (FC) layer every neuron of previous layer is connected to every neuron to the next layer in the model like normal multi-layer perceptron (MLP) model. This layer uses (always?) Softmax activation function in the output layer. FC layer takes the high level feature of previous layers as input and produces an N dimensional feature vector as output, where N is the number of output classes. Each value of the output vector represents probability of the class.

Loss Function:

Loss function is an important statistical parameter to measure the performance of the model, which is defined as the difference between predicted value and actual value. In order to increase the performance of the model loss should be minimized. Several loss functions are used to measure the performance of the network viz. L1 loss, L2 loss, cross entropy loss, Hinge loss etc. Cross entropy is the most preferred

loss function which is used in CNN. In multi-class classification problem multi-class cross entropy is used.

Regularization:

Regularization is a set of techniques which aims to reduce the overfitting problem of the trained model. Several regularization techniques are used in CNN to generalize the model during

classification:

1. L1 and L2 regularization

2. Dropout

3. Data augmentation etc.

We have implemented two different deep learning models based on modified CNN for the purpose of colour image recognition. The first model is basically an improvement over a previous CNN model (described in [3], whereas the second one is the newly proposed CNN model for this problem. In this first model, we have made two major changes. We have used Adam optimizer instead of using Adadelta optimizer. Also, to avoid overfitting in any circumstances, we have increased dropout percentage in between the second pooling layer and flatten layer by 15 percent.

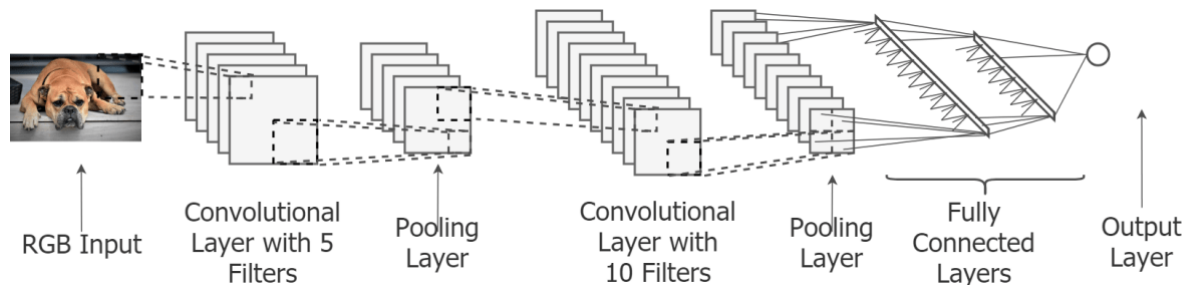


Fig.3. Detailed pipeline of proposed CNN model used for colour image recognition.

The second model is also based on CNN which has the following layers as shown in Figure 3. In the first convolutional layer, we have taken 32 learnable convolutional filters each with a kernel size of 55 pixels that act as a sliding window spatially of size 55 over all the images of size 2020 pixels and computed the respective dot product. Each filter produces a separate 2-dimensional activation map and these activation maps are stacked along the depth dimension (1 for our monochrome images) to produce the output volume. The size of the output images of convolution layers remain the same as we have used 'zero padding'. The output is then passed through ReLU activation function to avoid the negative values [8]. To reduce computational cost and to extract higher level features, we have then added pooling layers. We have used max pooling filters that go as a sliding window of size 22 over all the images of 2020 and the highest pixel value of every sliding window represents the window. Using max pooling the output image size has reduced to 1010 pixels, thereby reducing the image size by 75%.

In the second convolutional layer, we have taken 16 learnable filters each with a kernel size of 33 pixels, followed by ReLU activation. This layer is also followed by a max pooling layer of window size 22. A 30% dropout layer is then added in order to reduce overfitting [9]. The next layer is the flattened layer where we have converted the resulting 2-dimension output into a 1-dimensional vector in order to fit it into a FC layer. In the next phase, we have added two FC or dense layers with 128 neurons each having full connection to all the activations in the previous layers. In the following layer, we have excluded 50% neurons by using dropout. Finally, the dense layer containing 10 neurons for the 10 classes is added followed by Softmax function, which returns a list of probability corresponding to each of the 10 classes. The class with the highest probability value is taken as the final output.

5.Results and Analysis:

In this approach, we finally got the optimal accuracy 93.12% of an SVM classifier (with RBF kernel and $C = 10000$), which is impressive. To clearly understand how and why CNN works, we see here an insight into the internal operation and behavior of complex models. To understand what each layer has learned from the image, we need to determine what kind of input feature enables the map. Any feature map is enabled if its pixel value is 1. (Zeiger) [10] proposed a visualization technique using multi-layered deconvolutional network (DeconvNet) [11] to express the input stimuli that excited the individual feature map at any level of the model. A deconvnet can be thought of as a convent model that uses the same elements (filtering, pooling) but vice versa, thus reversing the pixels instead of mapping them to the properties. To test a conformant, the deconvnet is connected to each of its layers, providing a seamless way to return to the image Pixels. To test a given CovNet activation, we set all other level activity to zero and pass the feature maps as input to the connected deconvnet layer. Then we reconstruct the operation in the bottom layer until the input pixel space is found. Please see [10] [11] for more information about Deconvnet.

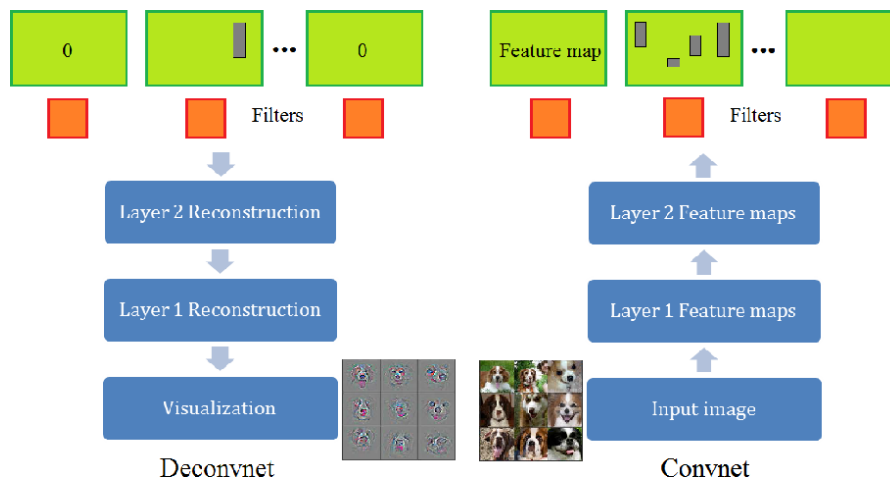


Fig:4. Projecting back from higher layers.

To test a given CovNet activation, we set all other level activity to zero and pass the feature maps as input to the connected deconvnet layer. Then we reconstruct the operation in the bottom layer until the input pixel space is found. Please see [10] [11] for more information about Deconvnet. Shows feature visualization for different levels [10]. Instead of showing a single strong activation for a given feature map, we show the top 9 activities. Each individually projected in the bottom pixel space reveals various structures that excite a given feature map, so its input is displayed in the deformation. In addition to these scenes we see related image patches. These have more variations than visualization because the focus is completely on the contrasting structure within each patch. For example, in layer 5, row 1, col 2, the patches appear to have little in common, but the visualizations reveal that this particular feature map focuses on the grass in the background, not the foreground objects. The projections from each layer show the hierarchical nature of the features in the network. Level 2 responds to angles and other edge / color combinations. Level 3 has more complex aggregation, capturing similar textures or patterns. Layer 4 shows significant variation, but it is more classified: dog face, bird's foot. Level 5 shows the whole object with significant pose variation, e.g., dogs and grass.

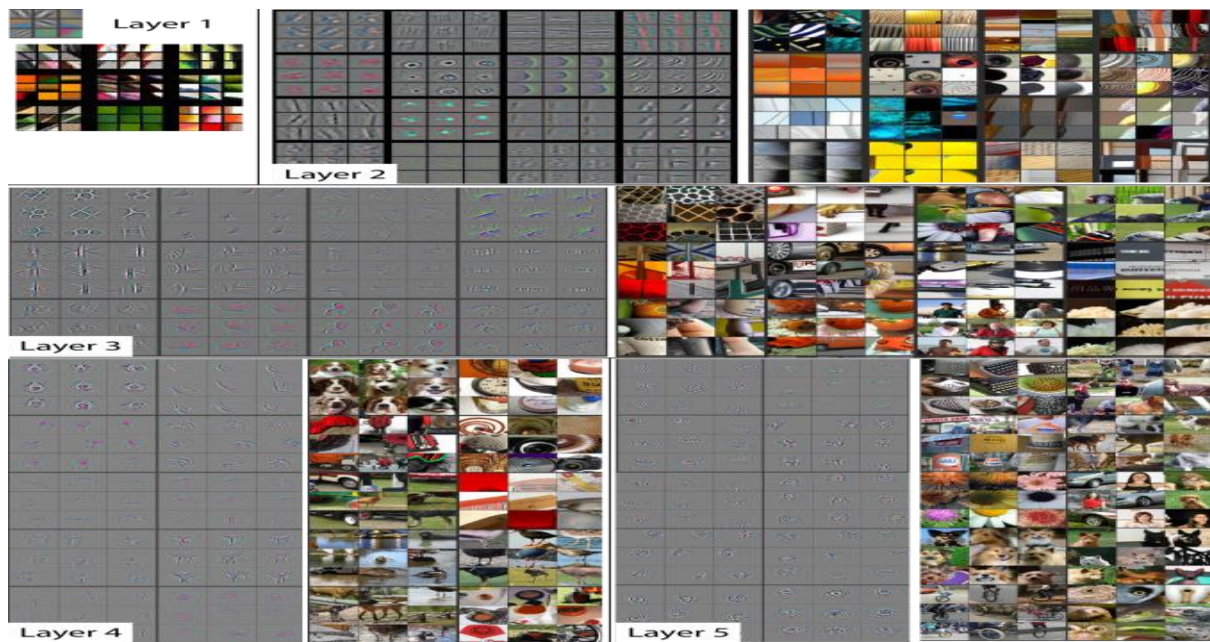


Fig :5. Some incorrectly classified images in the kaggle test set

Now we get an insight into why deep architectures can achieve better performance. Features learned by Deep Neural Networks are hierarchical. With more hidden layers, the learned features become more high level and specific. They are more expressive, class-specific and invasive in the background when compared to human-made features such as dense-shift or color features. We can also figure out which layers are critical to recognition and change the parameters and architecture of deep neural networks , e.g. increase the number of feature maps in some layers, to achieve better performance. To know the remaining issues and get better performance, we searched for what kind of image was misclassified. Figure 5 reveals some characters in the image that failed to classify. Some image resolutions (1 and 2) are too small to recognize, and some image critical characters, e.g. The face, is hidden (3 and 4).

The test set contains some cartoon images (5 and 6) that are just the general size of a dog or cat, which is hard to classify. Also, many misclassified images have backgrounds (7 and 8) that are very complex or similar to nozzle animals. Until the end, some images may not even be recognized by humans (9). From the features of the images that failed to be identified, we can try different techniques. We can try object localization to detect the animals in the images, so as to eliminate complex backgrounds. We can also take advantage of additional training sets to improve the efficiency of classifying special images such as cartoon dogs or cats. We can also combine features such as texture features with features learned in the Deep Neural Network, which can be helpful only in recognizing the fur or partial images of the animal's body.

Results

To achieve better performance, we applied our second approach, which is a trainable model that applies CNN to learn features. We saw insights into what Deep Networks has learned from the images and explain why they achieve good performance. This method has the highest accuracy of 93.12% (from an SVM classification), which helps us achieve the best results. In the case of categorization, we take into account our high dimensional feature space, mainly considering SVM and BP neural networks. Various parameter settings were searched to improve the accuracy of the classification in the test dataset. For example, for BP neural networks, we tried different latent layers and secret units; For the SVMs, different kernel functions and Python parameters were used. Table 1 illustrates the best results for each model and related parameters.

Table 1: Best Performance on the Test Dataset for Different Models

Feature	Classifier	Parameter Setting	Accuracy
Dense-SIFT	SVM	Linear kernel, C=1000	67.60%
Dense-SIFT+Colors	SVM	Linear kernel, C=5	71.47%
From Deep CNN	BP Network	1 Hidden layer, 30 Neurons	93.01%
From Deep CNN	SVM	RBF kernel, C = 10000	93.12%

In the future, we will explore further to achieve better performance. For example, we will try to change the architecture and parameter settings of the Deep Neural Network based on the feature view of the feature map at different levels. Also, we will apply different parameter settings for SVM and Deep Neural Networks. We can also try object localization to eliminate the effects of complex backgrounds. Additionally, we want to extract more features or try a combination of human-crafted features and learned features.

Conclusion:

In this paper, we have explored two CNN modified models for colour image recognition in this paper. We also have developed a new dataset to be used for image recognition, consisting of 10,000 images of size 2020 pixels each. Out of the two models, the proposed model has achieved a significant performance increase with a combination of dropout regularization, hyper parameters tuning and using Adam optimizer. In our dataset, we have obtained a benchmark classification accuracy of 93.12% by the proposed model. It may be stated that as CNN can be used to find unique patterns within the images so the results are far more promising than compared to the conventional machine learning approaches. Though the results are encouraging still there are some scopes for further improvement of the present work in terms of deep learning architecture and hyper parameter optimization. In the future, we will look forward to explore more sophisticated architectures along with expanding the application domain to image recognition and object identification.

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